

Casio OLED Screen Assembly Manual

Assembly Manual

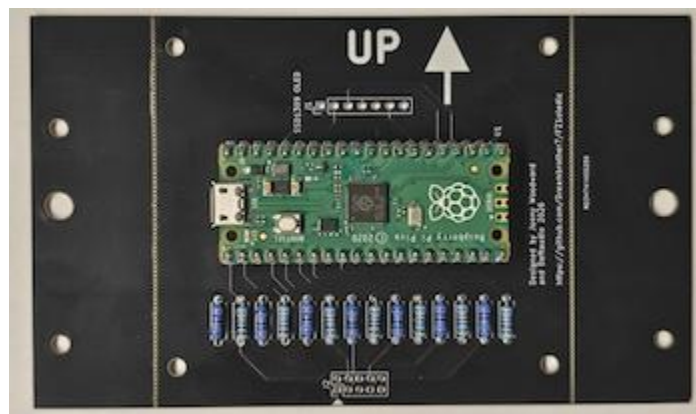
This kit is based on the recent development by Johny Woodward (aka Dreambrother7)
<https://github.com/Dreambrother7/FZ1oledlc>

Deftaudio designed a set of PCBs to support this development.
<https://github.com/Deftaudio/Casio-FZ-1>

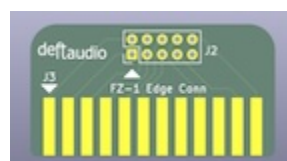
This product is developed as an open-source project. We believe the best designs are built collaboratively and welcome contributions from individuals and organizations alike.

The replacement OLED screen is meant to be universal across FZ-1, FZ-10m, FZ-20m, VZ-10m(starting Rev1.1) and VZ-1 (still needs testing). The communication protocol for those synthesizers is exactly the same, however the screen assembly in FZ-1 is much larger than in rack units, has different dimensions and a flex cable that connects it to the CPU board.

The base PCB assembly hosts Pico and an OLED screen. It connects to the adapter board that goes into the flex cable slot on Casio. The base PCB has snap off regions to be mounted in FZ-1, FZ-10m/FZ-20m or VZ-10m(Rev 1.1).



The adapter board is unique for FZ-1 or FZ-10m/FZ-20m/VZ-10m due to a different connector type which is used on the rack family. It connects with a 10-pin ribbon cable (2.00mm pitch).



BOM:

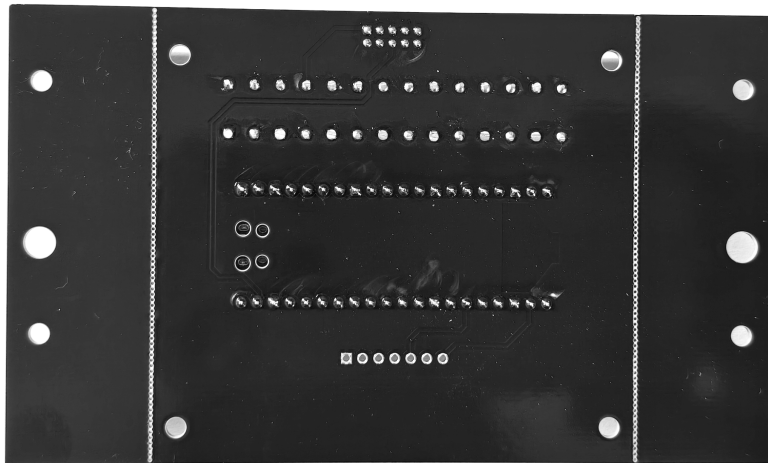
	Location	P/N
PCBs		
	OLED baseboard	
	Adapter board for FZ-1 or FZ-10m/FZ-20m/VZ-10m	
ICs		
	Raspberry Pi Pico 1	
Resistors		
	R1, R3, R5, R7, R9, R11, R13, R15	2k or 20k
	R2, R4, R6, R8, R10, R12, R14, R16	1k or 10k
Misc		
	OLED screen assembly with a header	SSD1309 compatible
	40-pin male header 2.54mm pitch	
	4pcs x 5-pin 2.0 male headers	
	2pcs x 10-pin female header	
	10-pin ribbon cable	

Assembly instructions.

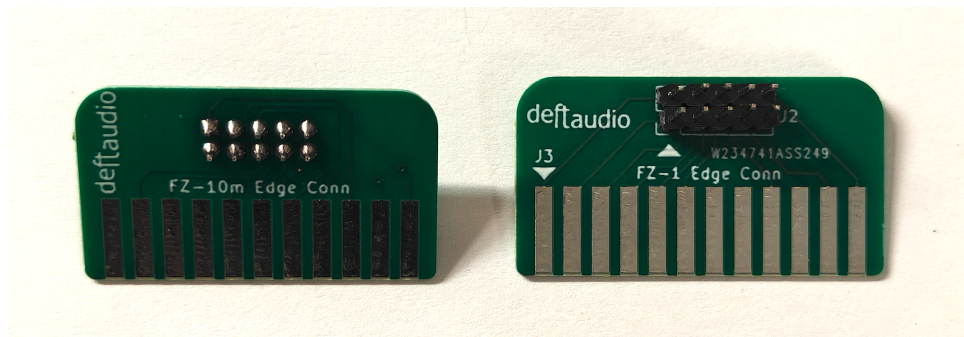
Solder resistors first. There are only two types of resistors used. Their main purpose is a voltage divider to adopt 5V to 3.3V logical inputs on Pico1. They have to be 2:1 ratio. The original design used 2k:1k, but it's more common to see 20k:10k since those drives less power and for Casio bus speeds are totally reasonable.

Then solder a male header into the Pico1 slot and solder Pico1 to it. The main reason for using a header is giving extra space. Otherwise the USB cable won't fit. It might be needed later, in case of any firmware updates from Johnny Woodward later. All Pico1 are shipped preprogrammed and you can solder it straight to the board if you don't plan to touch it again.

Next important step is to isolate all contacts on the board before installing a display, as its metal frame and components can cause a circuit short. Just use an electrical tape (in Deftaudio assembled units we're using 3M VFB double-side tape to secure the screen better)

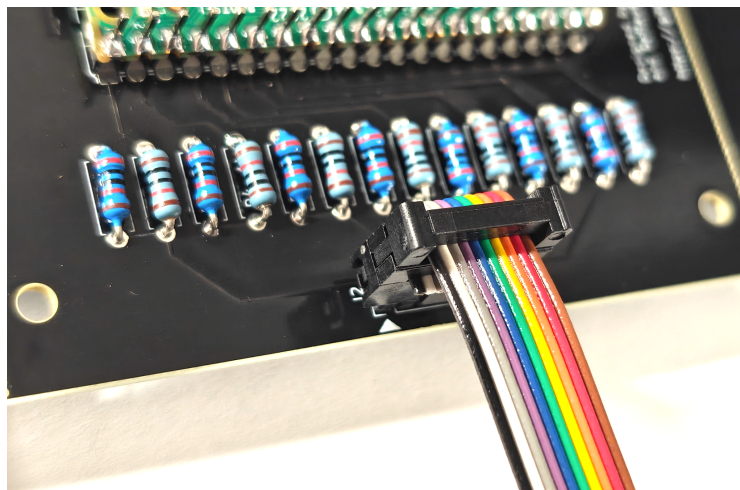


Now solder connector in adapter boards:

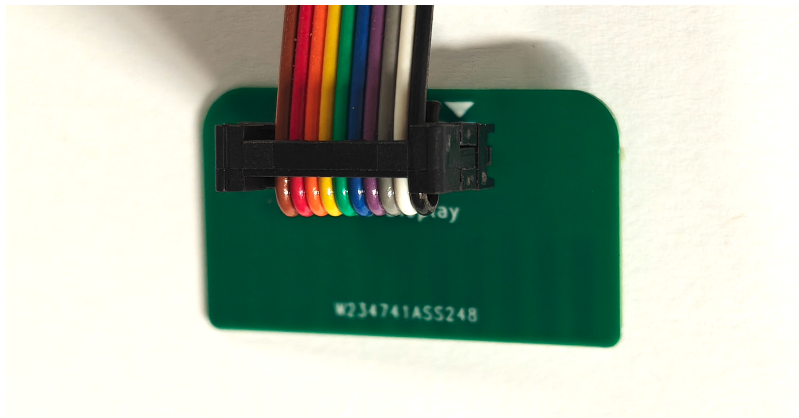


Please, note, the 10 pin connector is soldered in front of connector pins for FZ-1 adapter and on the back for FZ-10m/FZ-20m/VZ-10m adapters. This is important for the pinout and location of sockets.

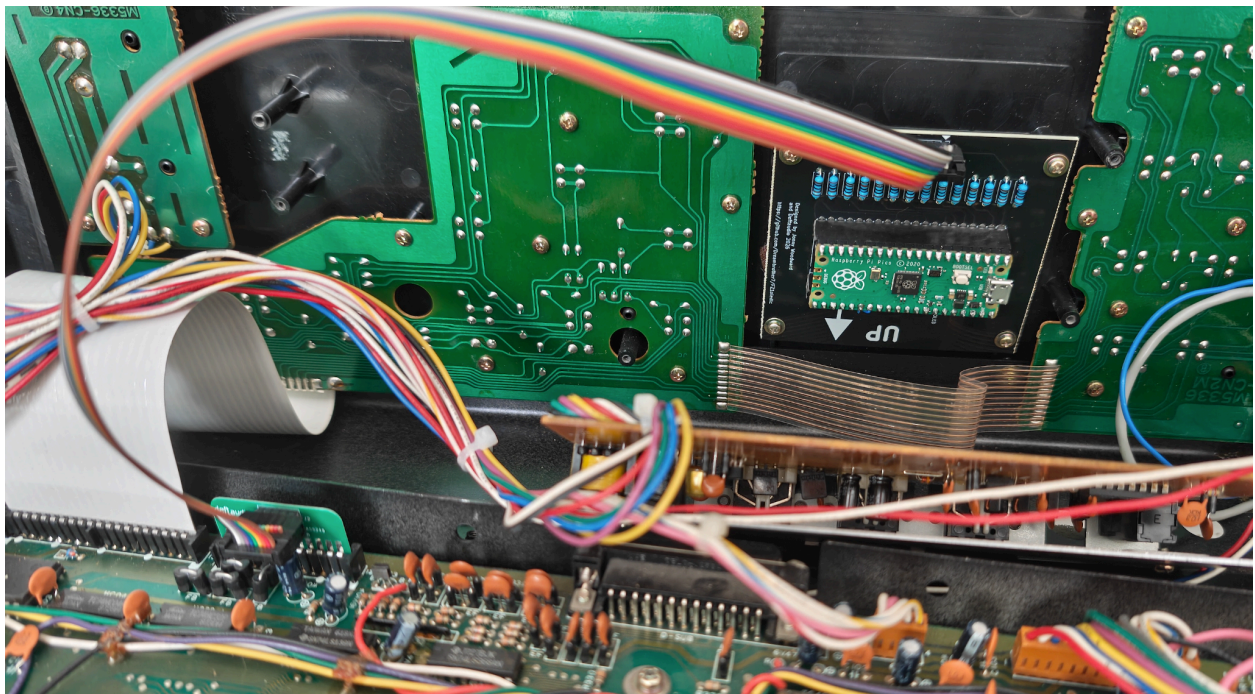
Make the cables. Follow Pin1 orientation (black wire) to correspond to the arrow on the connector. This then should match the arrow (pin1) on the PCB silkscreen



You should see the same arrow on the adapter board as well. Needless to say that reverting pins will cause Pico1 damage.



Installing into Casio is straightforward. Below FZ-1 is shown. You can see the board with sides removed. You may want to secure the adapter board with a hot glue, but usually the original connector holds it very well.



Troubleshooting:

Check voltages, cables, etc - perform some basic troubleshooting first.

For testing the OLED module itself, disconnect the adapter board, and upload a test firmware to Pico1. It allows just to test the screen and pico while powered through USB cable.

https://github.com/Dreambrother7/FZ1oledlc/releases/download/v0.2.0/FZ1_oled_test_defaudio_pcb_pico.uf2 Revert to normal firmware if the test is successful.